01.This checks the possible moves available from the current location and EV range.

```
102 ?- successors(state(colombo, 220), Moves).
Moves = [succ(state(kadawatha, 208), 'Drive colombo -> kadawatha (12km, 10.3 min)', 10.285
714285714286), succ(state(ratnapura, 119), 'Drive colombo -> ratnapura (101km, 86.6 min)',
 86.57142857142857), succ(state(galle, 101), 'Drive colombo -> galle (119km, 102.0 min)',
102.0)].
```

Why - This demonstrates the possible next destinations, distances, estimated travel times, and remaining EV range.

02. We use this command to compare how the three search algorithms find a route between the same two cities.

```
103 ?- compare_algorithms(colombo, kandy).

=== EV Trip: colombo -> kandy (battery 220 km full charge) ===

BFS (fewest actions):
    Drive colombo -> kadawatha (12km, 10.3 min)
    Drive kadawatha -> kegalle (65km, 55.7 min)
    Drive kegalle -> mawanella (20km, 17.1 min)
    Drive mawanella -> kandy (26km, 22.3 min)
  TOTAL: 105.4 min | 4 actions | 0 charge stops | 24 nodes expanded | 6.399 ms

DFS (first feasible route):
    Drive colombo -> kadawatha (12km, 10.3 min)
    Drive kadawatha -> kegalle (65km, 55.7 min)
    Drive kegalle -> mawanella (20km, 17.1 min)
    Drive mawanella -> kandy (26km, 22.3 min)
  TOTAL: 105.4 min | 4 actions | 0 charge stops | 5 nodes expanded | 0.289 ms

A* (minimum total trip time):
    Drive colombo -> kadawatha (12km, 10.3 min)
    Drive kadawatha -> kegalle (65km, 55.7 min)
    Drive kegalle -> mawanella (20km, 17.1 min)
    Drive mawanella -> kandy (26km, 22.3 min)
  TOTAL: 105.4 min | 4 actions | 0 charge stops | 7 nodes expanded | 16.504 ms
true.
```

Why: This lets you demonstrate the performance of BFS, DFS, and A* for the same starting and destination cities. You can compare the resulting routes, travel times, and expanded states if your predicate reports them.

03. "A* uses the cost of the route so far and an estimated cost to the destination to guide its search.

```
104 ?- run_astar(colombo, kandy).

A* (minimum total trip time):
    Drive colombo -> kadawatha (12km, 10.3 min)
    Drive kadawatha -> kegalle (65km, 55.7 min)
    Drive kegalle -> mawanella (20km, 17.1 min)
    Drive mawanella -> kandy (26km, 22.3 min)
  TOTAL: 105.4 min | 4 actions | 0 charge stops | 7 nodes expanded | 0.535 ms
true.
```

Why: It gives you a chance to explain A* specifically and how it uses path costs and a heuristic to guide the search.

04. We temporarily block a road and run the route search again to check whether the system finds an alternative route.

```
105 ?- block_road(kegalle, mawanella).
Road kegalle - mawanella is now BLOCKED.
true.

106 ?- compare_algorithms(colombo, kandy).

=== EV Trip: colombo -> kandy (battery 220 km full charge) ===

BFS (fewest actions):
    Drive colombo -> ratnapura (101km, 86.6 min)
    Charge at ratnapura (+101km, 50.5 min)
    Drive ratnapura -> nuwaraeliya (65km, 55.7 min)
    Drive nuwaraeliya -> kandy (77km, 66.0 min)
  TOTAL: 258.8 min | 4 actions | 1 charge stops | 31 nodes expanded | 1.956 ms

DFS (first feasible route):
    Drive colombo -> kadawatha (12km, 10.3 min)
    Drive kadawatha -> kegalle (65km, 55.7 min)
    Drive kegalle -> kadawatha (65km, 55.7 min)
    Drive kadawatha -> kegalle (65km, 55.7 min)
    Charge at kegalle (+207km, 69.0 min)
    Drive kegalle -> kadawatha (65km, 55.7 min)
    Drive kadawatha -> colombo (12km, 10.3 min)
    Drive colombo -> kadawatha (12km, 10.3 min)
    Drive kadawatha -> colombo (12km, 10.3 min)
    Drive colombo -> ratnapura (101km, 86.6 min)
    Charge at ratnapura (+202km, 101.0 min)
    Drive ratnapura -> nuwaraeliya (65km, 55.7 min)
    Drive nuwaraeliya -> kandy (77km, 66.0 min)
  TOTAL: 642.3 min | 13 actions | 2 charge stops | 14 nodes expanded | 0.881 ms

A* (minimum total trip time):
    Drive colombo -> ratnapura (101km, 86.6 min)
    Charge at ratnapura (+101km, 50.5 min)
    Drive ratnapura -> nuwaraeliya (65km, 55.7 min)
    Drive nuwaraeliya -> kandy (77km, 66.0 min)
  TOTAL: 258.8 min | 4 actions | 1 charge stops | 58 nodes expanded | 3.547 ms
true.

107 ?- unblock_road(kegalle, mawanella).
Road kegalle - mawanella is now OPEN.
true.
```

- Why: This demonstrates that your route planner can handle a road restriction and that you can restore the road afterward.

05. We temporarily disable a charging station and check how the route planner responds.

```
108 ?- block_station(kegalle).
Charger at kegalle is now OUT OF SERVICE.
true.

109 ?- compare_algorithms(colombo, kandy).

=== EV Trip: colombo -> kandy (battery 220 km full charge) ===

BFS (fewest actions):
    Drive colombo -> kadawatha (12km, 10.3 min)
    Drive kadawatha -> kegalle (65km, 55.7 min)
    Drive kegalle -> mawanella (20km, 17.1 min)
    Drive mawanella -> kandy (26km, 22.3 min)
  TOTAL: 105.4 min | 4 actions | 0 charge stops | 23 nodes expanded | 1.924 ms

DFS (first feasible route):
    Drive colombo -> kadawatha (12km, 10.3 min)
    Drive kadawatha -> kegalle (65km, 55.7 min)
    Drive kegalle -> mawanella (20km, 17.1 min)
    Drive mawanella -> kandy (26km, 22.3 min)
  TOTAL: 105.4 min | 4 actions | 0 charge stops | 5 nodes expanded | 0.397 ms

A* (minimum total trip time):
    Drive colombo -> kadawatha (12km, 10.3 min)
    Drive kadawatha -> kegalle (65km, 55.7 min)
    Drive kegalle -> mawanella (20km, 17.1 min)
    Drive mawanella -> kandy (26km, 22.3 min)
  TOTAL: 105.4 min | 4 actions | 0 charge stops | 7 nodes expanded | 0.456 ms
true.
```

- Why-This shows the EV-specific feature of handling an unavailable charging station.